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Article

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Posted Date: June 11th, 2026

DOI: <https://doi.org/10.21203/rs.3.rs-9845242/v1>

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Additional Declarations: No competing interests reported.

Design and Modeling of a Fractional-Order Fuzzy Logic Controller for BLDC Motor Drives in Electric Tractor Systems

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ABSTRACT

These Days Electric tractors are becoming more popular among farmers due to their environmental friendliness. The efficiency, torque responsiveness, and control stability of these systems could be compromised due to the dynamic performance of the Brushless DC (BLDC) motors used. This could happen under different loads and field situations. In order to improve the dynamic performance of electric tractors driven by BLDC motors, this paper presents a FO-FLC, which stands for Fractional Order Fuzzy Logic Controller. Combining the robustness of fractional calculus with the flexibility of fuzzy logic, the proposed controller outperforms PI and integer-order fuzzy controllers in terms of control accuracy and system stability. The FOFLC optimizes its steady-state and transient behavior with operators for fractional-order differentiation and integration, and it can make real-time adjustments to its control settings with fuzzy inference rules. We used MATLAB/Simulink simulations to assess its speed regulation, torque ripple mitigation, and transient response to load disturbance capabilities. When compared to conventional control systems, the FOFLC achieves far better results in terms of torque smoothness, overshoot reduction, and settling time. These findings show that electric tractors powered by FOFLC-based BLDC motor control can improve energy efficiency, drivability, and operational reliability in the future.

Keywords: BLDC motor, Fractional Order Control, Fuzzy Logic Controller, Electric Tractor, Dynamic Performance, MATLAB/Simulink.

1 Introduction

A more environmentally friendly alternative to traditional diesel-powered tractors, electric tractors are quickly becoming the norm due to the growing worldwide concern for sustainable agriculture and the need to reduce carbon emissions. High efficiency, less noise, cheaper maintenance expense, and no emissions on the field are just a few of the benefits of electric tractors. Due to its high torque density, wide speed range, and good efficiency, the Brushless DC (BLDC) motor has become the favored choice among numerous motor types appropriate for traction applications. Although BLDC motors are commonly used in agricultural settings, it is still difficult to provide accurate control over their speed and torque due to their nonlinear features, parameter fluctuations, and load disturbances[14].

Optimal performance under such changing situations is typically lost by traditional control systems like Proportional-Integral (PI) controllers because of their set parameters and lack of adaptability[24]. The introduction of intelligent control strategies such as Fuzzy Logic Control (FLC) has helped to overcome these constraints by making systems more adaptable and robust. Without a precise mathematical model of the system being required, fuzzy controllers are able to successfully manage uncertainty by mimicking human reasoning. The dynamic tuning capabilities of traditional FLCs is generally limited, and they might not deliver the best transient performance[2]. Fractional Order Control (FOC), a notion derived from fractional calculus, has emerged as a potent instrument for improving control robustness and flexibility in recent times. Fractional order controllers provide an extra tuning and dynamic adjustment degree of freedom by using non-integer differentiation and integration[23]. The Fractional Order Fuzzy Logic Controller (FO-FLC) is an amalgam of fractional order control and fuzzy logic that combines the best features of both approaches, allowing for accurate dynamic adjustment and intelligent decision-making[32]. Electric tractors, which can replace traditional diesel-powered tractors, have been developed at a rapid pace in response to the growing need for environmentally friendly farming methods. Zero emissions, great economy, less

noise, and reduced running expenses are just a few of the benefits of electric tractors[26]. But how well they work is mostly dependent on the powertrain, especially the electric motor and the way it is controlled. Because of its wide speed range, efficiency, minimal maintenance requirements, and high torque-to-weight ratio, the Brushless DC (BLDC) motor has become the favored choice among diverse motor types[5, 17, 19, 33]. Nonlinear dynamics and parameter variations due to back-EMF distortion, temperature changes, and load fluctuations are observed in BLDC motors during field operations, notwithstanding these advantages. When it comes to electric tractors, these problems are made worse by the fact that agricultural jobs like plowing, tilling, and transporting include mechanical loads with a high degree of inertia, frequent torque transients, and traction demands that vary rapidly. As a result, consistent functioning in different field conditions, fast dynamic response, and minimum torque ripple are all achieved with a robust control technique[11]. The speed and torque of BLDC motors have historically been controlled by Proportional-Integral (PI) controllers because of their simplicity and ease of implementation[6]. On the other hand, traditional PI controllers frequently underperform when faced with nonlinear and time-varying situations. Their inflexible gains make them ill-suited for the fast load changes that are common in tractor operations, leading to overshoot, slowed settling time, and diminished performance[16]. One solution to these problems is the introduction of Fuzzy Logic Controllers (FLCs), which provide adaptive control by simulating human thinking through rule-based decision-making. By efficiently addressing nonlinearities and uncertainties without necessitating an explicit mathematical model, FLCs strengthen the stability of BLDC motor control[13]. However, FLCs' performance is still susceptible to constraints imposed by their rigid structure as well as the adjustment of scaling factors and membership functions. Fractional-Order Control (FOC) methods have become popular in recent years due to their capacity to introduce a new degree of freedom by means of fractional integral and derivative orders, which are not integers. The system dynamics can be better controlled and made more resilient throughout a wide frequency range thanks to this flexibility. The Fractional-Order Fuzzy Logic Controller (FOFLC) is an amalgam of Fuzzy Logic and Fractional-Order Control that brings together the best features of both methods, including robustness, flexibility, and dynamic performance that has been fine-tuned[7]. When it comes to electric tractors, this controller is ideal because of how quickly it can adjust to changing field loads and how smoothly it can distribute torque. The FOFLC is able to manage nonlinearities efficiently, decrease overshoot, increase transient response, and boost the BLDC motor's overall dynamic performance. As a result, electric tractors become more reliable, efficient, and productive[27].

1.1 Problem Statement

The effectiveness and durability of the BLDC motor control system determine the dynamic performance of electric tractors, which are a sustainable solution for modern agriculture. Despite their widespread use, traditional PI controllers are ill-equipped to deal with the nonlinear dynamics, high inertia, and rapidly changing load circumstances seen in transportation, plowing, and tilling, among other field operations. These controllers have poor stability, slow transient response, torque ripple, and speed oscillations under varying loads and speeds because they are not very adaptable. Despite their longevity and flexibility, Fuzzy Logic Controllers (FLCs) struggle to handle the wide dynamic range of electric tractor applications due to their membership functions and fixed rule bases. The fractional differentiation and integration orders of fractional-order controllers (FOCs) allow for more tuning freedom, which in turn improves system response. It may be difficult for independent FOC designs to adjust settings and respond rapidly to changing operating conditions. To enhance the dynamic performance, torque ripple, and robustness of electric tractors driven by BLDC motors, a hybrid control method is required. This technique should incorporate both fuzzy logic's adaptive reasoning and fractional calculus' fine-tuning flexibility. The Fractional-Order Fuzzy Logic Controller (FOFLC) is a solution to these issues; it improves the operational performance, reliability, and efficiency of electric tractors by ensuring stable operation, rapid response, and smooth control of torque under different field and load conditions. The goal of this project is to enhance the dynamics of electric tractors by designing and implementing a BLDC motor drive system that is based on FO-FLC. Under varying loads, the proposed controller lessens speed fluctuations, flattens torque, and quickens transient response. Using MATLAB/Simulink simulations, the performance of FO-FLC is compared to that of integer-order fuzzy controllers and classic PI controllers. A potential solution for advanced electric tractor applications, the proposed technology improves speed response, stability, and efficiency.

2 Literature Survey

Here are some papers related to fractional order fuzzy logic BLDC motor control, and what they contribute, plus gaps or trade-offs. The Some of identified contribution and gap from the literature review shown in Table 1.

Because of its widespread use in electric vehicles, robotics, and renewable energy systems, Brushless DC (BLDC) motor control has been the subject of much research. Smooth torque delivery, reliable operation under fluctuating field loads, and good energy efficiency are all dependent on the dynamic performance of BLDC drives in electric tractors. To improve the performance of BLDC motors, numerous researchers have looked into various control systems, each with their own set of pros and cons[20]. Because of their simplicity and ease of implementation, traditional Proportional-Integral (PI) controllers have seen widespread use. But in nonlinear and time-varying systems, their adaptability is limited by their set gain parameters. As

Table 1. Identified research gaps from the literature review

Paper / Title	Proposed Method and Findings	Identified Gaps	Aspects Not Addressed
Eltoum et al. [22]	Combines a fuzzy logic controller for reference current management with a FOPID controller for inverter or DC-bus voltage regulation. Control parameters are optimized using a modified harmony search algorithm.	Achieves reduced deviation, lower torque ripple, improved settling time, and smaller steady-state error compared to standalone FOPID or fuzzy controllers.	Designed and tuned for pre-defined load and speed profiles. May not effectively handle severe load fluctuations or highly dynamic traction loads encountered in electric tractors. Practical implementation constraints such as computation delay and power limitations are not comprehensively addressed.
S. Das et al. [21]	Fuzzy logic dynamically tunes fractional-order PI gains. Conventional PI, FOPI, and fuzzy-based controllers are evaluated under multiple operating modes including constant speed, constant torque, and motoring/generating conditions.	Significant improvements in rise time, overshoot, peak response, and torque–speed tracking across operating modes.	Real-time tractor operating conditions such as abrupt load spikes, terrain variations, and slope changes are not considered. Hardware validation is not demonstrated, limiting practical applicability in agricultural vehicles.
Putta Edukondalu et al. [29]	A hybrid control scheme where a fuzzy logic controller regulates the DC-bus voltage and a FOPID controller manages the reference current. Parameters are optimized using harmony search and evaluated at different speeds and load conditions.	Demonstrates superior robustness and performance under varying speeds and load conditions.	Electric tractor-specific dynamics such as vehicle inertia, battery aging effects, heavy-duty traction loads, gearbox dynamics, and mechanical linkages are not incorporated into the model.
I. Podlubny et al. [25]	Demonstrates that fractional-order fuzzy logic controllers outperform conventional PI controllers in dynamic renewable energy systems, providing faster recovery and improved transient behavior.	Establishes the superiority of fractional-order and fuzzy-based control strategies over integer-order controllers in dynamic environments.	The study is not focused on BLDC motor drives for electric tractors, which exhibit distinct mechanical inertia, load variability, and control architecture requirements.

an example, it was shown in [15] that traditional PI control works well at steady-state but has problems with overshoot and slow settling time when the load changes. Because of this restriction, researchers have been looking into more sophisticated and intelligent control techniques. A lot of people are looking to Fuzzy Logic Controllers (FLCs) to deal with nonlinearities and changes in parameters. By using rule-based reasoning, FLCs are able to successfully deal with system uncertainties without requiring an accurate mathematical model. When compared to conventional PI control in BLDC motor drives, fuzzy-based control improves transient responsiveness and increases resilience [8, 15]. While basic FLCs are generally effective, their performance can be affected by poorly designed membership functions and rule bases. A new and exciting method for controlling motors has been Fractional Order Control (FOC) in the last several years. Additional tuning flexibility and enhanced system robustness to parameter variations are introduced by FOC by permitting the order of differentiation and integration to be fractional rather than integer. Fractional Order PI (FOPI) controllers were initially proposed by Podlubny[4], who showed that they might enhance phase and gain margins. Subsequently, FOPI control was included in BLDC drives, as reported in [30], leading to more consistent torque and quicker speed regulation. Expanding on these developments, the Fractional Order Fuzzy Logic Controller (FOFLC) demonstrates better performance and adaptability in complicated nonlinear systems by combining Fractional Order Control with Fuzzy Logic. Research by authors like [30] and [18] shows that when compared to traditional FLC and FOPI controllers, FO-FLC performs better in terms of settling time, stability margins, and overshoot reduction. Fuzzy logic's heuristic decision-making power and fractional calculus's exact tuning flexibility come together in its hybrid form. Research on implementing FOFLC in electric tractor drives has been limited, despite great advances in intelligent control systems for BLDC motors. This is because electric tractor drives face important obstacles such as changing terrain resistance, frequent load disturbances, and efficiency optimization. Thus, in order to address this knowledge gap, this paper will design and evaluate a fractional order fuzzy logic controller for electric tractors powered by BLDC motors, with an emphasis on enhancing both the dynamic performance and the overall stability of the drive.

2.1 Research Gap Identified

While there has been significant progress in electric vehicle and motor control technologies, there are still some limitations in controlling the BLDC motor drives for electric tractor applications. The conventional PI/PID controllers used in BLDC motor drives are not very robust in the presence of unpredictable agricultural loads, nonlinear disturbances and uncertain parameters. Although current fuzzy logic controllers are more adaptive, they tend to have slower transient responses, limited dynamic precision, and trouble with memory-dependent nonlinear behaviors, all of which are problems in electric tractor operations. Most of the reported BLDC motor control algorithms for EVs focus on typical urban EV applications rather than addressing low-speed, high-torque agricultural electric tractor settings. However, the agricultural electric traction systems that use fractional-order controllers have not been studied enough in terms of how to combine them with fuzzy logic techniques, even though the controllers are more robust and flexible. Current research rarely deals with the simultaneous minimization of torque ripple, improvement of energy efficiency, rejection of load disturbance and smooth speed regulation. The current research lacks the comprehensive modeling framework to evaluate the fractional-order fuzzy controllers under real tractor operating conditions, such as uneven terrain, unexpected load variations and field traction requirements.

2.2 Novelty of the Proposed Work

The proposed work introduces a new Fractional-Order Fuzzy Logic Controller (FO-FLC) for BLDC motor drives for electric tractor systems. The most significant innovations are: Enhancing the dynamic efficiency and robustness of BLDC motor drives through the development of a hybrid control framework combining FLC and fractional calculus. Introducing fractional order tuning parameters to provide adaptive control and higher precision, to give more freedom than conventional integer order fuzzy controllers. Designing the controller with high start-up torque, variable field loads and low-speed operation of electric tractors in mind. Improving energy economy and speed tracking performance, while minimizing overshoot, settling time and steady-state error. Elaboration of the full mathematical and computational model of the BLDC drive systems of electric tractors taking into account the real farming conditions. The objective of this study is to show that our fuzzy logic controller is more stable and disturbance-rejecting than conventional PI and classical fuzzy logic controllers.

3 Methodology

This section outlines the proposed Fractional Order Fuzzy Logic Controller (FO-FLC) design, its integration with the BLDC motor drive system, and the overall approach followed to improve the dynamic performance for electric tractor applications. The methodology consists of system modeling, controller design, and simulation analysis in MATLAB/Simulink.

3.1 System Overview

The proposed system focuses on the speed control of a BLDC motor used in an electric tractor drive train. The overall control architecture integrates the FOFLC with the electronic commutation and inverter stage of the BLDC motor[10].

3.1.1 BLDC Motor Model

An accurate representation of the BLDC motor's dynamic behavior is achieved by the use of electrical characteristics, back EMF, and torque in the model. Figure 1 shows a system that uses a FOFLC to precisely control the speed of a BLDC motor in an electric tractor. The procedure starts by converting a three-phase alternating current (AC) source to direct current (DC) using a rectifier [12]. The voltage supplied to the motor can be regulated by modulating this DC supply using a PWM rectifier. A shaft encoder reads the motor speed ω_s and reports it to the control system along with the actual speed ω_m). To determine the inaccuracy, the FOFLC evaluates the real speed in relation to the reference speed (ω_{ref}). It is the rate of change of this inaccuracy that the FLC uses to modify the reference current (i_{ref}). In order to operate the motor, the reference current generator takes this current and produces the correct phase currents (i_{aref} , i_{bref} , i_{cref}) [31].

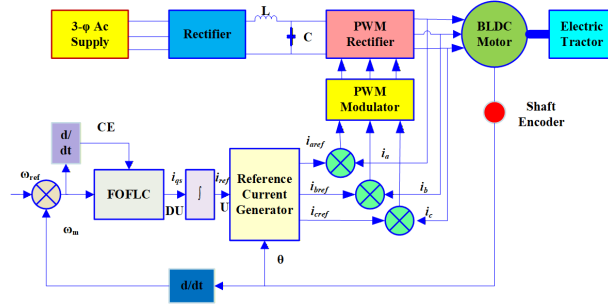


Figure 1. The block diagram of FOFLC -based speed control of BLDC motor

- The reference speed is compared with the actual speed obtained from the motor feedback sensor.
- The error (e) and change in error (Δe) are fed into the FO-FLC, which processes them through a fuzzy inference system combined with fractional-order operators.
- The FO-FLC generates a control output that adjusts the duty cycle of the Pulse Width Modulation(PWM) signals.
- These PWM signals drive the inverter, controlling the stator voltage applied to the BLDC motor.
- The controller continuously adjusts the voltage to maintain desired speed and torque even under load variations.

3.2 Modelling of BLDC Motor

The BLDCM rotor is basically a permanent magnet with three stator windings. Because magnets and stainless steel have a high resistance, rotor currents can be ignored. The damper winding simulation does not exist [1]. The phase variables for each of the three windings are determined by calculating the circuit equation.

$$\begin{bmatrix} V_{rst} \\ V_{yst} \\ V_{bst} \end{bmatrix} = \begin{bmatrix} R_{st} & 0 & 0 \\ 0 & R_{st} & 0 \\ 0 & 0 & R_{st} \end{bmatrix} \begin{bmatrix} i_{rst} \\ i_{yst} \\ i_{bst} \end{bmatrix} + \frac{d}{dt} \begin{bmatrix} L_{rr} & L_{ry} & L_{yb} \\ L_{yr} & L_{yy} & L_{yb} \\ L_{br} & L_{by} & L_{bb} \end{bmatrix} \begin{bmatrix} i_{rst} \\ i_{yst} \\ i_{bst} \end{bmatrix} + \begin{bmatrix} e_r \\ e_y \\ e_b \end{bmatrix} \quad (1)$$

where v_{ast} , v_{bst} , and v_{cst} denote stator phase voltages, and R_s denotes stator resistance. The three-phase currents of the stator are denoted as i_{cst} , i_{ast} , and i_{bst} . The self-inductance levels for each phase are denoted as L_{aa} , L_{bb} , and L_{cc} . E_c , E_b , and E_a are linked to electromotive pressures, and phase-to-phase inductance are L_{ac} , L_{bc} , and L_{ab} . Every winding's resistance has been taken for granted [9]. Furthermore, it is thought that the rotor reluctance stays constant at any angle because the rotor is not visible.

$$L_{rr} = L_{yy} = L_{bb} = L \quad (2)$$

$$L_{ry} = L_{yr} = L_{rb} = L_{br} = L_{yb} = L_{by} = M \quad (3)$$

The PMBLDC model is constructed by replacing equation (1) with equations (2) and (3).

$$\begin{bmatrix} V_{rst} \\ V_{yst} \\ V_{bst} \end{bmatrix} = \begin{bmatrix} R_{st} & 0 & 0 \\ 0 & R_{st} & 0 \\ 0 & 0 & R_{st} \end{bmatrix} \begin{bmatrix} i_{rst} \\ i_{yst} \\ i_{bst} \end{bmatrix} + \frac{d}{dt} \begin{bmatrix} L & M & M \\ M & L & M \\ M & M & L \end{bmatrix} \begin{bmatrix} i_{rst} \\ i_{yst} \\ i_{bst} \end{bmatrix} + \begin{bmatrix} e_r \\ e_y \\ e_b \end{bmatrix} \quad (4)$$

where v_{ast} , v_{bst} , and v_{cst} source voltages.

$$v_{rst} = v_{ro} - v_{no}, v_{yst} = v_{yo} - v_{no} \text{ and } v_{bst} = v_{bo} - v_{no} \quad (5)$$

where v_{no} is the zero-reference voltage at the dc connection and v_{ro} , v_{yo} , v_{bo} , and v_{no} are neutral voltages.

The stator phase currents are limited to be balanced i.e.

$$i_{rst} + i_{yst} + i_{bst} = 0 \quad (6)$$

from the above equation it can easily understand the source inductance..

$$M_{iy} + M_{ib} = -M_{ir} \quad (7)$$

As a result, during the state space domain,

$$\begin{aligned} \begin{bmatrix} V_{ast} \\ V_{bst} \\ V_{cst} \end{bmatrix} &= \begin{bmatrix} R_{st} & 0 & 0 \\ 0 & R_{st} & 0 \\ 0 & 0 & R_{st} \end{bmatrix} \begin{bmatrix} i_{ast} \\ i_{bst} \\ i_{cst} \end{bmatrix} \\ &+ \frac{d}{dt} \left(\begin{bmatrix} L-M & 0 & 0 \\ 0 & L-M & 0 \\ 0 & 0 & L-M \end{bmatrix} \begin{bmatrix} i_{ast} \\ i_{bst} \\ i_{cst} \end{bmatrix} \right) \\ &+ \begin{bmatrix} e_a \\ e_b \\ e_c \end{bmatrix} \end{aligned} \quad (8)$$

Before, it was thought that the back EMFs (e_r , e_y , and e_b) exhibited a trapezoidal wave coming from

$$\begin{bmatrix} e_r \\ e_y \\ e_b \end{bmatrix} = \omega_m \lambda_m \begin{bmatrix} f_{rs}(\theta_r) \\ f_{ys}(\theta_r) \\ f_{bs}(\theta_r) \end{bmatrix} \quad (9)$$

In this context, ω_m represents the angular rotor speed in rad/sec, where m is the flux linkage and r is the rotor position in radians. The functions $f_{rs}(\theta_r)$, $f_{ys}(\theta_r)$, and $f_{bs}(\theta_r)$ are equivalent to e_r , e_y , and e_b , respectively, up to a maximum magnitude of ± 1 . Induced emfs don't have sharp corners because of their trapezoidal shape. The electromagnetic fields, which are a continuous function, are generated by the derivatives of flux connections [9]. Because of its fringes, the flux density function is smooth and devoid of sharp edges. The electromagnetic torque, according to Newton, is

$$T_e = e_r i_r + e_y i_y + e_b i_b \quad (10)$$

The inertia is defined as

$$J = J_m + J_l \quad (11)$$

The following equation includes the variables inertia (J), friction coefficient (B), and load torque T_l :

$$J \frac{d\omega_m}{dt} + B\omega_m = (T_e - T_l) \quad (12)$$

The speed-position relationship of the electrical rotor

$$\frac{\lambda_m}{J} f_{bs}(\theta_r) \quad (13)$$

The system is nonetheless affected by the damping coefficient B, even if it is usually relatively tiny. The variable " θ_r " stands for the rotor position in the previous equation, which is iterated every 2ϕ cycles. To avoid a voltage imbalance and maintain the drive efficiency, it is necessary to take into account the potential of the point of neutrality in relation to zero potential (v_{no}) [28]. Equation (6) can be substituted for equation (8) in the VI equation to obtain.

$$\begin{aligned} v_{r0} + v_{y0} + v_{b0} - 3v_{no} &= R_s (i_r + i_y + i_b) \\ &+ (L-M) (p i_r + p i_y + p i_b) \\ &+ (e_r + e_y + e_b) \end{aligned} \quad (14)$$

Changing equation (14) to equation (6) yields the following

$$v_{r0} + v_{y0} + v_{b0} - 3v_{n0} = (e_r + e_y + e_b) \quad (15)$$

Thus

$$v_{n0} = \frac{v_{r0} + v_{y0} + v_{b0} - (e_r + e_y + e_b)}{3} \quad (16)$$

Equations (8), (12), and (13), which are differential equations, are related. explains the model using variables that are independent i_{rst} , i_{yst} , i_{bst} , ω_m , and θ_r . The following equation is the result of combining all applicable equations:

$$\dot{x} = Ax + Bu + Ce \quad (17)$$

Where

$$x = [i_{rst} \quad i_{yst} \quad i_{bst} \quad \omega_m \quad \theta_r]^t \quad (18)$$

$$A = \begin{bmatrix} -\frac{R_{st}}{L-M} & 0 & 0 & -\frac{\lambda_m J f_{rs}(\theta_r)}{P} & 0 \\ 0 & -\frac{R_{st}}{L-M} & 0 & -\frac{\lambda_m J f_{ys}(\theta_r)}{P} & 0 \\ 0 & 0 & -\frac{R_{st}}{L-M} & -\frac{\lambda_m J f_{bs}(\theta_r)}{P} & 0 \\ \frac{\lambda_m}{J} f_{rs}(\theta_r) & \frac{\lambda_m}{J} f_{ys}(\theta_r) & \frac{\lambda_m}{J} f_{bs}(\theta_r) & -\frac{B}{J} & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad (19)$$

$$B = \begin{bmatrix} -\frac{1}{L-M} & 0 & 0 & 0 & 0 \\ 0 & -\frac{1}{L-M} & 0 & 0 & 0 \\ 0 & 0 & -\frac{1}{L-M} & 0 & 0 \\ 0 & 0 & 0 & -\frac{1}{L-M} & 0 \end{bmatrix} \quad (20)$$

$$C = \begin{bmatrix} -\frac{1}{L-M} & 0 & 0 \\ 0 & -\frac{1}{L-M} & 0 \\ 0 & 0 & -\frac{1}{L-M} \\ 0 & 0 & 0 \end{bmatrix} \quad (21)$$

$$u = [v_{rst} \quad v_{yst} \quad v_{bst} \quad T_l]^t \quad (22)$$

$$e = [e_r \quad e_y \quad e_b]^t \quad (23)$$

Like an induction motor, a BLDC motor features a permanent-magnet rotor and a stator with three phases of windings. The tool's back electromagnetic field is trapezoidal. To manage the conduction's switching sequence, the inverter acts as an electronic commutator [7]. You need to know where the BLDC motor's rotor is in order to change its speed. Hall effect sensors are one tool for this kind of learning. Three Hall Effect sensors, 120 degrees apart, make up a typical BLDCM arrangement. The sensor always displays a HIGH logic state whenever the rotor passes it. The LOW logic state will be retained if this does not happen. Every sensor has finished one full cycle by the time a cycle ends [27]. According to Figure 2, the phase voltage changes depending on the states of the hall sensors, and the back EMF that is produced has a trapezoidal form. The rotor is spun by the stator coils, which are powered by Hall effect sensors that sense its position. The stator and rotor coils of the BLDCM are shown in Figure 3. With each rotation of sixty degrees, the data is read from the hall effect sensors. A complete cycle consists of six distinct stages [28]. Once started, the motor continues to turn by means of a series current flowing from the two poles that are opposite to each other. There are two coils that are simultaneously carrying current. The three-phase inverter converts the direct current (DC) power from the tractor's battery to three-phase alternating current (AC) power, which is then sent to the BLDC motor. The PWM modulator transforms the reference currents into switching signals that power the BLDC motor on the tractor. As part of a closed-loop system, the motor runs smoothly and effectively at the specified speed thanks to constant input that adjusts for variations in load and outside influences. The use of hall sensors to ascertain the rotor's location guarantees precise electrical commutation. That Block in the FO-FLC Controller The control signal is utilized to regulate the voltage and torque of the motor, with the speed error and change in error taken into consideration. The Load Model is a useful tool for simulating the electric tractor's operating conditions in the field, including the weight of the load it is pulling.

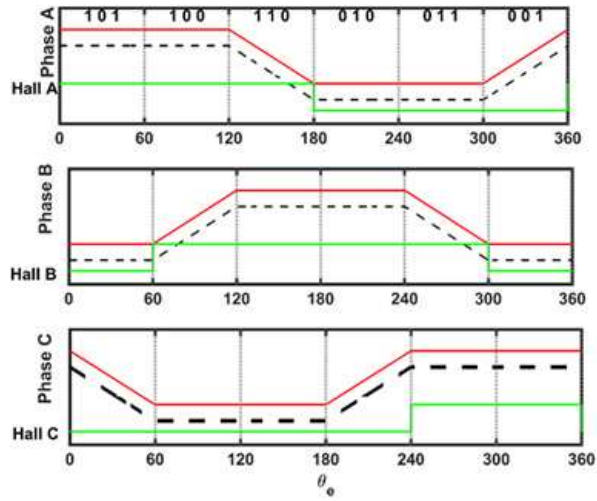


Figure 2. Phase and hall sensor signals

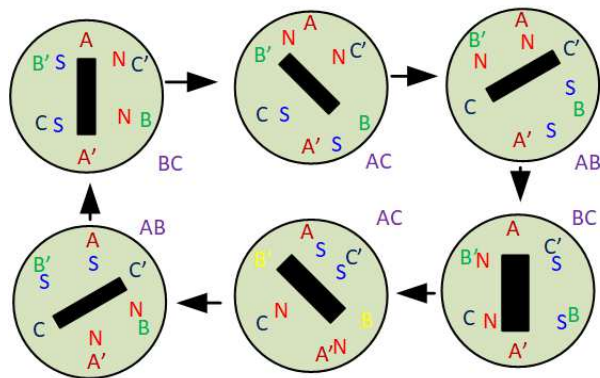


Figure 3. Commutation process of BLDCM

4 Control method

The methodology of this research involves the modeling, design, simulation, and performance evaluation of a Fractional-Order Fuzzy Logic Controller (FOFLC) for the BLDC motor drive used in electric tractor applications.

4.1 Fractional-Order Operators

The fractional-order elements are implemented using the recursive approximation, which approximates the fractional integral and derivative terms within the frequency range of interest. The proposed Fractional Order Fuzzy Logic Controller (FOFLC) algorithm governs the BLDC motor speed by intelligently adjusting the inverters PWM signal based on the deviation between the desired and actual motor speed. The algorithm integrates fuzzy reasoning for decision-making and fractional calculus for fine-tuning system dynamics.

4.1.1 Design Of Filter Using Fractional Order Controller

With the development of the FOC (Fractional Order Controller), different definitions of FODs and FOIs emerged, as discussed in [3]. A very efficient method of dealing with the problems is to use filters. According to [22], there are a few sequential filters. We can write the fractional order function λ using the Fractional Order Filter (FOF) if we assume the expected fitting range is $(\omega_b; \omega_h)$.

$$F(s) = \left[\frac{1 + \frac{as}{b\omega_a}}{1 + \frac{bs}{b\omega_h}} \right]^\lambda \quad (24)$$

Where $0 \leq \lambda \leq 1$; $s = j\omega$, $a \geq 0$; $b \geq 0$, and

$$F(s) = \left(\frac{as}{b\omega_a} \right)^\lambda \left(1 + \frac{-bs + b}{bs^2 + a\omega_h s} \right)^\lambda \quad (25)$$

In the frequency range " $\omega_b \geq \omega \geq \omega_h$ ", by using a "Taylor- series" expansion, we obtain

$$F(s) = \left(\frac{as}{b\omega_a} \right)^\lambda \left(1 + (s) + \frac{\lambda(\lambda-1)}{2} P^2(s) \dots \right) \quad (26)$$

Where

$$F(s) = \left(1 + \frac{-bs + b}{bs^2 + a\omega_h s} \right) \quad (27)$$

It is then found that

$$s^\lambda = \frac{(b\omega_a)^\lambda a^\lambda}{a^\lambda \left(1 + (s) + \frac{\lambda(\lambda-1)}{2} P^2(s) \dots \right)} \left(\frac{1 + as/b\omega_a}{1 + bs/b\omega_b} \right)^\lambda \quad (28)$$

from this the "Taylor-series" to leads to

$$s^\lambda = \frac{(b\omega_a)^\lambda a^\lambda}{a^\lambda (1 + \lambda P(s))} \left(\frac{1 + as/b\omega_a}{1 + bs/b\omega_b} \right)^\lambda \quad (29)$$

Thus, the FOD is characterized as

$$s^\lambda \approx \left(\frac{b\omega_a}{b} \right)^\lambda \left(\frac{ds^2 + b\omega_h s}{d(1-\lambda)s^2 + b\omega_h s + d\lambda} \right) \left(\frac{1 + bs/d\omega_b}{1 + ds/d\omega_h} \right)^\lambda \quad (30)$$

When every pole lies on the left side of the complex s-plane, we get an equation that is equal (30). In the previous equation, the two extremes are

1. one pole must be located at $-a\omega_h/b$ which is -Ve Real Pole since $\omega_h \geq 0, a \geq 0, b \geq 0$;
2. two other poles must be the roots of the equation

$$b(1-\lambda)s^2 + a\omega_h s + b\lambda \quad (31)$$

Consequently, Equation (30) has stable poles in the interval (ω_l, ω_h) . The repeated-time coherent model provides an approximation for the irrational fractional-order section of the expression in (29).

$$F(s) = \lim_{N \rightarrow \infty} F_N(s) = \lim_{N \rightarrow \infty} \prod_F^N = -N \frac{1 + s/\omega'_k}{1 + s/\omega_k} \quad (32)$$

To express the zero and pole of rank k in terms of the algorithmic distribution of real numbers, we can say that they are

$$\omega'_k = \left(\frac{b\omega_a}{a}\right)^{\frac{\lambda-2k}{2N+1}}, \omega_k = \left(\frac{b\omega_b}{a}\right)^{\frac{\lambda+2k}{2N+1}} \quad (33)$$

therefore, the model can be expressed as

$$s^\lambda \approx \left(\frac{b\omega_a}{b}\right)^\lambda \left(\frac{ds^2 + b\omega_h s}{d(1-\lambda)s^2 + b\omega_h s + d\lambda}\right) \prod_F^N = -N \frac{1 + s/\omega'_k}{1 + s/\omega_k} \quad (34)$$

If $a = 12$ as well as $b = 10$, this analysis can triumph in capturing the core of good. The FOS might be the most preeminent number-order system if the estimating method is correct. Rotating the order sign (λ) allows a fractional order controller to get. Therefore, λ can be integrated within the range of 0 to 1. Figure 4 shows the FOPI-controller construction. Figure 4 shows the system's arrangement and design of the K_p element, λ , for controller design. Each of the K_p and K_i is custom-made and installed in the building according to the design below.

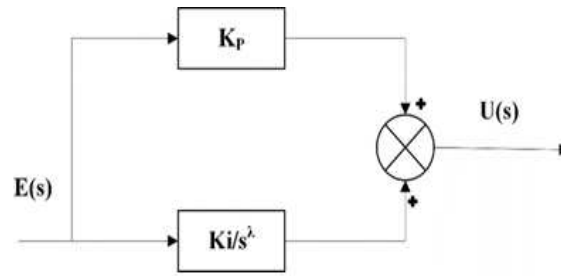


Figure 4. Scheme of FOPI controller.

4.2 Controller Design

- The implementation of PI, FLC, and FOPI controllers allows for performance comparison. Their findings will be used to measure how well the proposed FOFLC works. For increased control precision and flexibility, the suggested FOFLC combines fuzzy logic with fractional calculus. A error (e) and its change in error (de) are the two inputs that make up the controller. One result: a signal for control (u) In order to fine-tune the dynamic response more accurately than integer-order controllers, the FOFLC employs fractional-order integration (λ) and differentiation (μ) operations.
- Fuzzification: Converts input error and change in error into linguistic variables (e.g., Negative Big, Zero, Positive Small, etc.).
- Rule Base: A set of if-then? rules define control behaviour based on the relationship between inputs and output.

example rule:

- IF error is Positive Big AND change in error is Negative Small THEN output is Positive Medium.
- Defuzzification: Converts fuzzy output back to a crisp control signal using the centroid method.

4.2.1 Structure of FOFLC

The fundamental level of a Simple FLC is shown in Fig. 5. To construct a fuzzy set, the membership functions for input ω_{ref} and $\Delta\omega_{ref}$ are displayed in Figures 6 and 7, respectively. As follows. Figures 8 and 9 show the membership functions that were output, which are ΔK_P and ΔK_I , respectively. To improve the Mac-Vicar-Whelan inference table for strong FLC resilience, the FOFL is built as shown in Figure 10. The inputs are the error(E) and change in-error (CE), and the outputs are ΔK_P and ΔK_I .

4.2.2 Learning Part

With fuzzy rule base set theory, system designers have more leeway to communicate their observations using simple knowledge base notation. For FLC, closed-loop control systems with nonlinear i/p and o/p are ideal. Conventional controllers, like FOC, function by analyzing the values of K_p and K_i , which are gains, and by analyzing input errors. Hence, for complex, non-linear systems, the expected level of activity from the Controller is not met. A variable gain for K_p , full terms, and Derivatives might be employed in lieu of a static one. Improving controller performance and swiftly stabilizing system output in the face of variation and external disturbances are both achieved by dynamic gain adjustment in a FOPI control architecture. Considering these characteristics, this article suggests a strategy that combines FLC and FOPI. The FOFL controller combines the FLC and FOPI into one. As part of this controller design, the FLC is configured to take the system error and error derivative as inputs and calculate the proportional scale factor "E." The entire terms of these scale factors will be used to adjust the controller gain at each sample period. The fundamental structure of a standard FOFL control mechanism is shown in Figure 10. These data are utilized to update the gain parameters of the FOFL controller, since the FLC in the proposed control structure uses them to determine the scale factor for proportional and integral terms using the error and its derivative. The FOFL controller's K_p and K_i signal gain levels are calculated using the following formulas:

$$K_p = k_p + \Delta K_p \quad (35)$$

$$K_i = k_i + \Delta K_i \quad (36)$$

Where k_p and k_i are the Fixed gain values.

4.3 Algorithm Steps

- Start the System: Initialize all control parameters, reference speed (ω_{ref}), and fractional orders λ and μ .
- Measure Motor Speed Obtain the actual speed (ω_{actual}) from Hall effect sensors.
- Compute Speed Error: $e(t) = \omega_{ref} - \omega_{actual}$: This represents the deviation of the actual speed from the desired reference.
- Compute Change in Error: $\Delta e(t)$. It indicates how fast the error is changing with time.
- Fuzzification: Convert the numerical values of $e(t)$ $\Delta e(t)$ into linguistic variables using predefined fuzzy membership functions (NB, NM, NS, ZE, PS, PM, PB).
- Fuzzy Inference (Rule Base Evaluation): Apply fuzzy if-then rules to determine the control action. Example: If (error is NB) and (change in error is NS), then (output is NB).
- Defuzzification: Convert the fuzzy output into a crisp control signal (u) using the centroid method.
- Fractional Order Processing: Modify the control signal using fractional-order integration and differentiation: This enhances flexibility and stability by allowing non-integer order tuning.
- PWM Signal Generation: Use the processed control signal $u(t)$ to generate the PWM duty ratio for inverter switching.
- Apply to BLDC Motor Drive: The inverter drives the BLDC motor according to the PWM control signal, adjusting the stator voltage and torque.
- Feedback and Update: Measure the new motor speed, update $e(t)$ and $\Delta e(t)$, and repeat the control loop.
- Stop Condition: The loop continues until the system reaches steady-state speed or the tractor control command changes.

Figure 11. shows the FO-FLC control algorithm flowchart structure. The FO-FLC combines the advantages of fuzzy inference and fractional calculus. Its structure comprises three key stages:

- Both the speed error (e) and the change in error (Δe) are inputs to the fuzzy logic process. By utilizing triangle membership functions, each input is transformed into fuzzy linguistic variables, including Negative Big (NB), Negative Medium (NM), Negative Small (NS), Zero (ZE), Positive Small (PS), Positive Medium (PM) and Positive Big (PB).
- The control action is defined by a set of fuzzy if-then rules called inference mechanisms. For example, if mistake is NB and change in error is NS, then output is NB. Effective handling of nonlinear behavior and parameter fluctuations is included into the fuzzy rule basis. The centroid approach is used to defuzzify the output, transforming it back into a clear control signal. The inverter's PWM duty ratio is determined by this output.

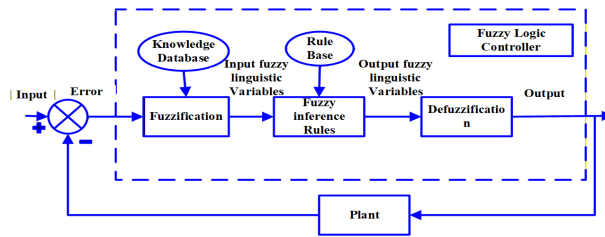


Figure 5. Structure of FL controller.

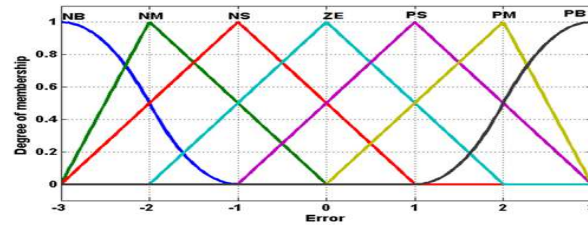


Figure 6. Membership function for error.

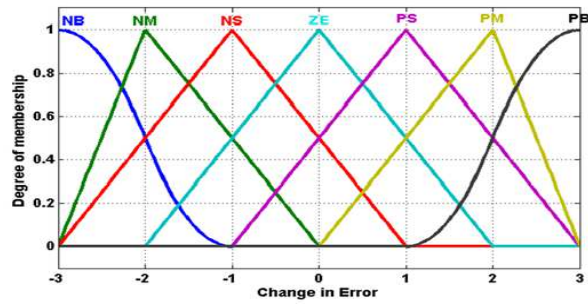


Figure 7. Membership function for change in Error .

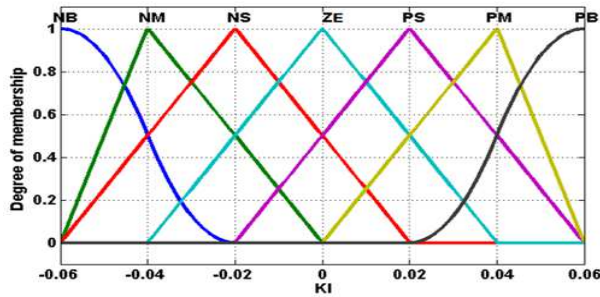


Figure 8. Membership function for ΔK_i .

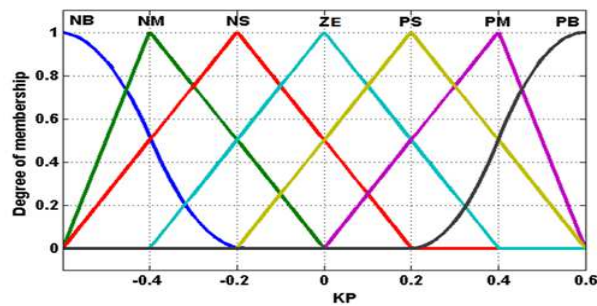


Figure 9. Membership function for ΔK_p .

- The fractional-order integrator and differentiator, denoted by fractional powers (integration order) and μ (differentiation order), are used to process the control output of the FLC. By providing more tuning options than the standard PI controller, this fractional dynamic adjustment improves control precision.

5 Results and Discussion

5.1 With different Load

Figure 12(a) displays No-load, 50% load and 100% load for the BLDC Motor, which is controlled by a fuzzy logic controller. The system quickly gets close to the reference speed with very little steady-state inaccuracy. The fuzzy logic controller works well when there isn't much mechanical stress. When the motor is working at 50% load, the overshoot is slightly more. Adding a load slows the speed down by 0.2 seconds. After this load, the fuzzy controller quickly brings the speed back to a level that is similar to the reference value. The fuzzy logic controller keeps the system stable even when the speed drops below the reference value, even when there is clear steady-state error. When the motor is working at 100% load, the overshoot is more. Adding a load slows the speed down by 0.2 seconds due to FLC it achieves near to the Reference speed and improves the rise time. Both the voltage and current flowing through the BLDC motor's stator are shown in Figure 12(b). Subsequent to 0.1 seconds, the current via the stator reaches a stable state. It is possible to see the BLDC motor's back electromotive force (emf) in its unloaded state in the waveform. There is a 100V setting for the 3-phase back emf voltages after 0.050 seconds, according to the result. Figure 13(a) shows the speed response of Different load conditions such as No-load, 50% and 100% load using Adaptive FLC. It observes that compared to Fuzzy logic controller, there is the improvement with respect to rise time, Settling Time and overshoot. During Load condition, the speed response is reached near to the Reference speed. Figure 13(b) shows the BLDC motor's stator voltage and current. At the one-second mark, the stator current has stabilized. Without a load, the BLDC motor's back electromotive force (emf) is shown by the waveform. Findings show that 100V is the value established for the 3-phase back emf voltages after 0.050 (secs). After what seems like only 0.2 seconds, the motor settles into a steady speed of 1000 rpm. Figure 14(a) shows how the motor drive system's speed responds when controlled by a Fractional-Order Fuzzy Logic Controller (FoFLC) under three different loads: no load, 50% load, and 100% load. The green horizontal line shows the reference speed, which is always 1000 RPM. The colored curves show the actual motor speed that was measured with the FoFLC. In the no-load situation, the motor speeds up quickly from a stop and reaches the reference speed in a short amount of time. There is a small overshoot at the start of the response, followed by behavior that is well-damped. With almost no steady-state error, the speed quickly settles at the reference value. This answer shows that the FoFLC has better dynamic performance in terms of Rise time and Settling time and Overshoot. The motor has a little more overshoot when it starts up under 50% load since it needs more torque. When the load is put on at about 0.2 seconds, the speed drops slightly. But the FoFLC quickly brings the speed back to the reference value. The motor has the most overshoot during the transient period when it is running at full load. When full load is suddenly applied at about 0.2s, the speed drops noticeably. Still, the FoFLC keeps the system stable and makes sure that recovery goes smoothly but very well regulated, showing that the system is more stable while it is under heavy load. With no load, half load, and full load, the motor drive system's response time to speed changes is shown in Figure 15. Three different types of fuzzy logic controllers, adaptive FLC and a FoFLC—are used to regulate the system. 1000 RPM is reference speed. All three controllers maintain a constant watch on the reference speed when the load is not present. With the least amount of overshoot, the quickest settling, and the shortest rise time, the FoFLC approaches the reference speed. There is a bit more overshoot before adaptive FLCs settle, but they are slightly quicker than regular FLCs. A sign of effective regulation occurs when there is no load if all controllers achieve little steady-state error. However, the FoFLC's superior transient performance demonstrates its superior responsiveness. The FoFLC has better performance during Transient. With its quicker settling time, less overshoot, and superior speed control under 100% loads, the FoFLC outperforms both controllers. The stability and strength of the FoFLC are unaffected by the fact that its speed decreases under load. The proposed controllers for BLDC motors are compared in Fig. 15 according to their rising time, settling time, and overshoot(%) performances. Based on the summary data from Table 2 for the controllers, FOFCL has the shortest rise time (0.025 seconds), settling time (0.085 seconds), and overshoot percentage (6%). According to the findings, the three-phase back emf voltages are set to 100V at the 0.001Sec mark. Adaptive FL controllers, FOFCL, PI, and fuzzy logic all contributed to the speed responses shown in Fig. 15. FOFCL has the shortest settling time (0.16 seconds) and the highest overshoot percentage (6%), as seen in the controller summary data in Table 3. At 50% load, the FOFCL outperformed the PI controllers, Fuzzy logic Controller, and Adaptive FLC. Table 3 shows the results of comparing the suggested controller to the dynamic performance of BLDC motors. Table 3 displays a comparison of the control performance and BLDC motor's performance at 50% load. In Table 4 you can see the specifications for the parameters. We compare it to conventional controllers and look at performance parameters including rise time, settling time, and overshoot.

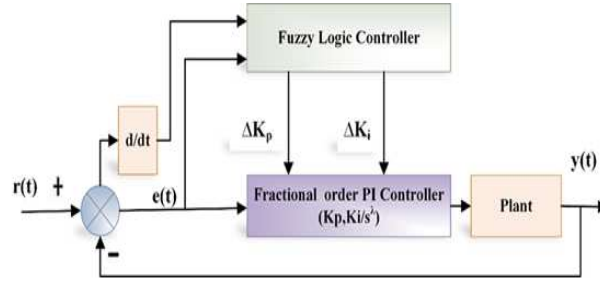


Figure 10. Block diagram of FOFL controller.

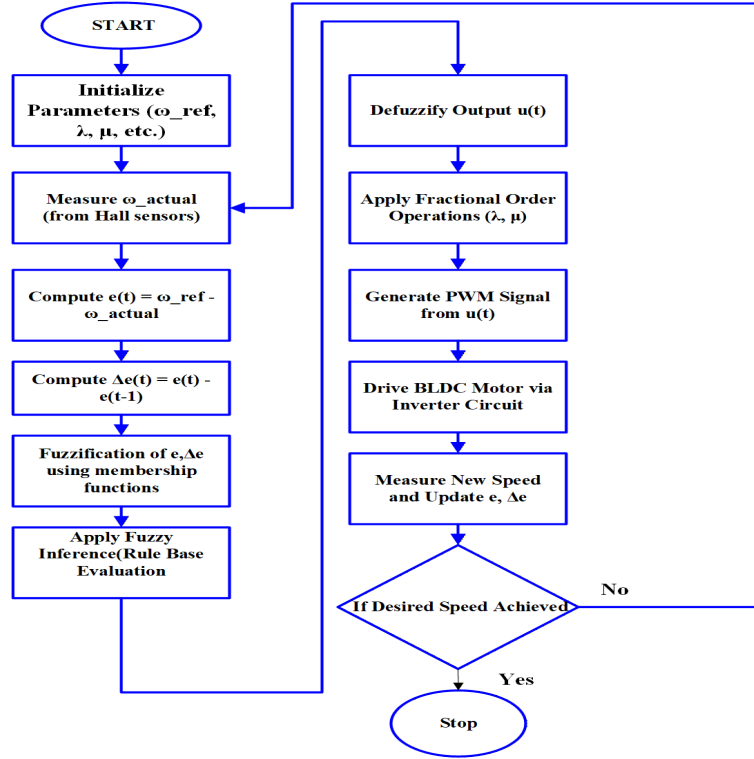


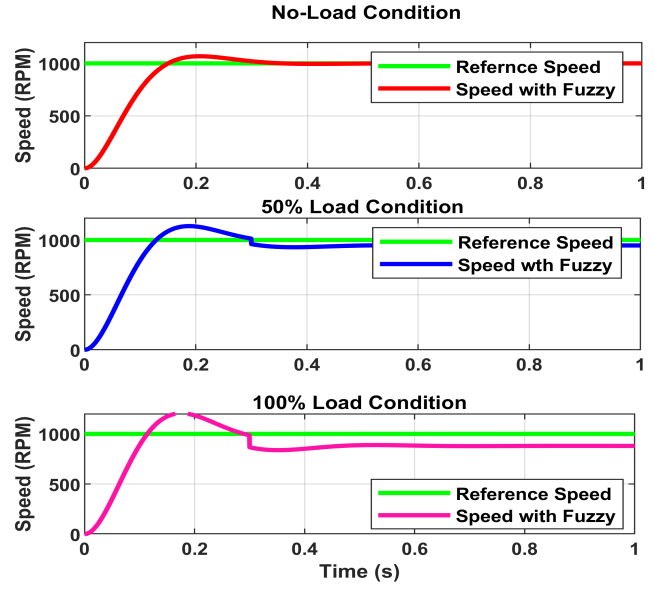
Figure 11. FO-FLC control algorithm flowchart structure.

Table 2. Comparisons of dynamic performance of BLDC Motor on No Load

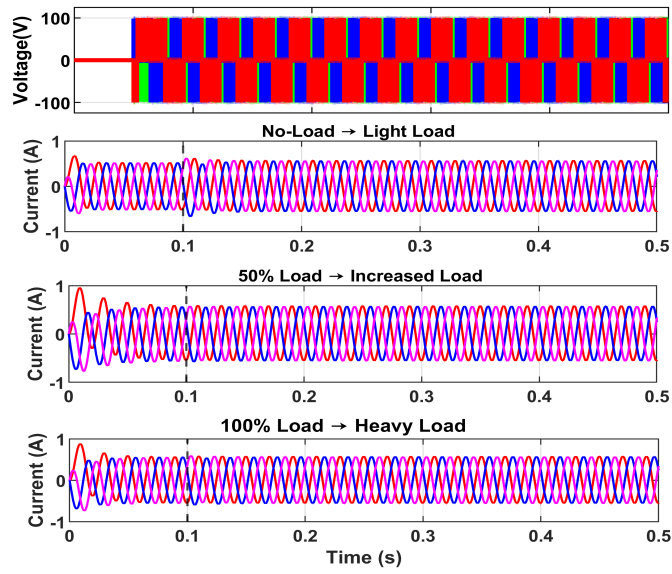
S.No	Controller	T_r (ms)	T_s (ms)	Overshoot(%)
1	With PI	80	850	15
2	With FLC	70	250	13
3	With adaptive FLC	30	100	8
4	With FOFLC	25	85	6

Table 3. Control performance comparison of BLDC Motor at 50% load

S.No	Controller	T_r (ms)	T_s (ms)	Overshoot(%)
1	With PI	320	400	10
2	With FLC	280	300	8
3	With Adaptive FLC	180	180	7
4	With FOFLC	150	160	6



(a)

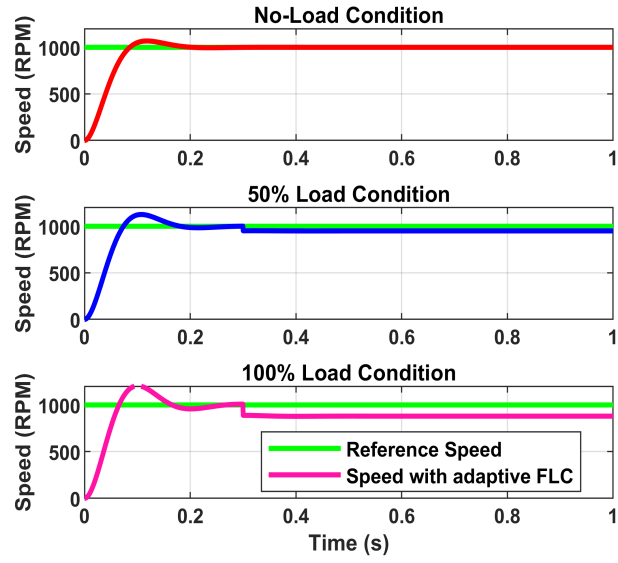


(b)

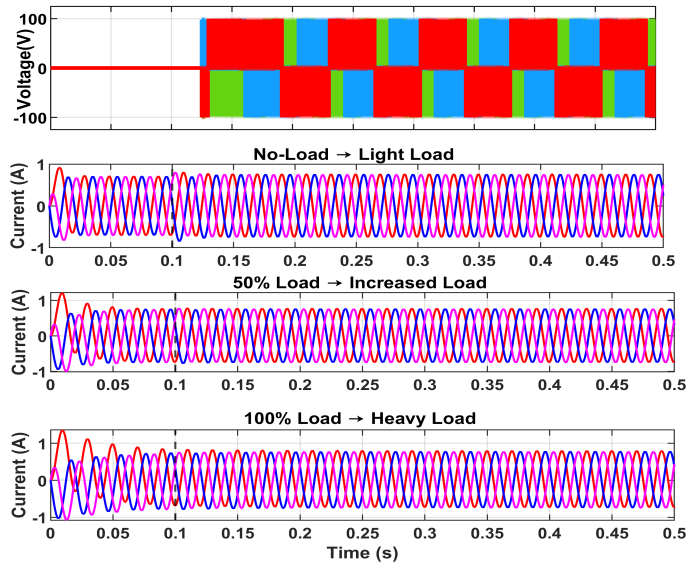
Figure 12. (a) Speed Response with FLC(b) Phase voltage and Phase current with FLC.

Table 4. Parameters Specifications

Parameter	Symbol	Value
DC Bus Voltage	V _{dc}	48 V
Motor Rated Power	P	1.5 kW
Rated Speed	ω_r	1000 rpm
Stator Resistance	R	1.2ohms
Stator Inductance	L	3.5 mH
Number of Poles	P	4
Fractional Orders	λ, μ	0.9,0.8

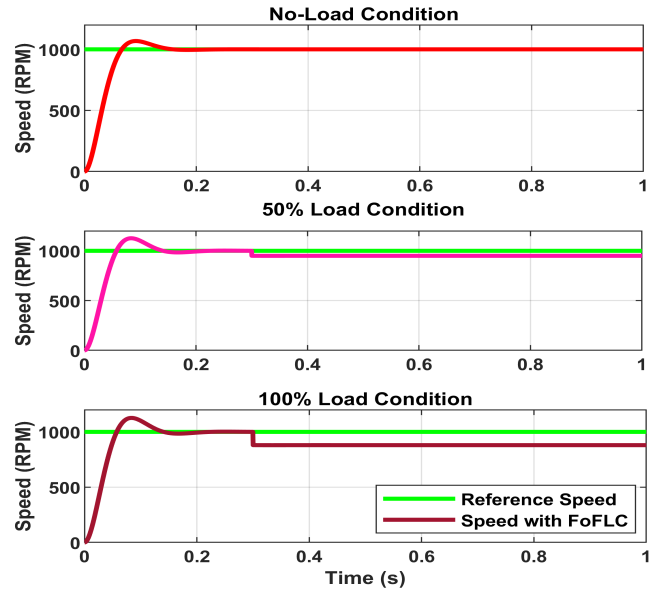


(a)

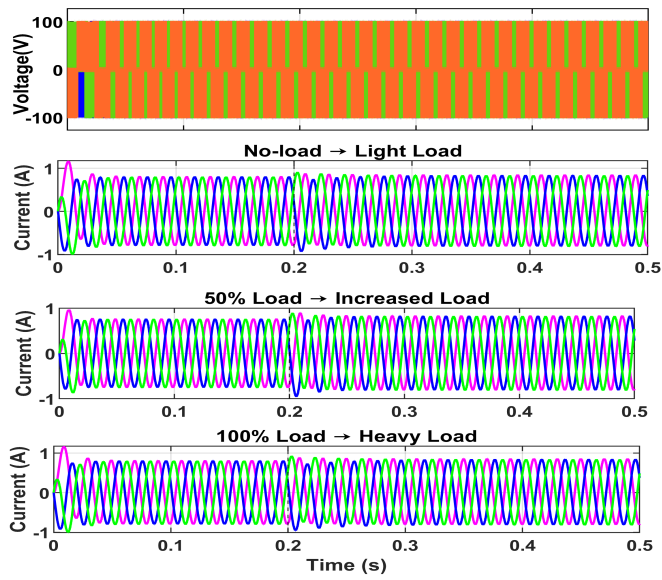


(b)

Figure 13. (a)Speed Response with adaptive FLC(b) Phase voltage and Phase current with adaptive FLC.



(a)



(b)

Figure 14. (a)Speed Response with FOFLC (b) Phase voltage and Phase current with FOFLC.

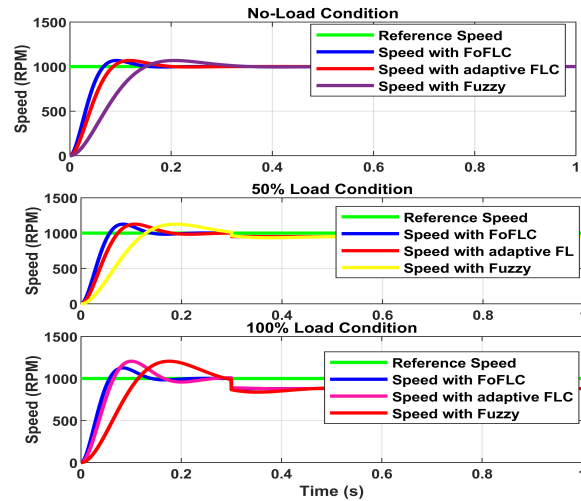


Figure 15. Variation of Speed with all Controllers.

6 Conclusion

This research employs MATLAB/Simulink modeling and simulation to assess the efficacy of a FOFCLC-based speed controller for a Permanent Magnet Brushless DC (PMBLDC) motor. A comprehensive model of the motor drive system was created and then tested under varying loads to determine how well the controller managed the speed. The outcomes are contrasted with the efficiency of FOFCLC, adaptive Fuzzy Logic Controller (AdFLC), fuzzy logic controller, and PI controller. At both no load and 50% of full load, the results showed that the FOFCLC controller worked far better than the adaptive FLC, FLC, and PI units. Its benefits were a shorter settling time, less overshoot, and a quicker response time overall. When it comes to operating PMBLDC motors in dynamic situations, FOFCLC is the way to go because it can stabilize the motor speed fast without sacrificing precision or efficiency. Under electric tractor-typically variable-load conditions, the suggested fractional-order fuzzy logic controller improves the BLDC motor's speed regulation and torque performance. Optimal transient and steady-state performance can be achieved more efficiently with the incorporation of FoFLC compared to traditional partial impulse (PI), fuzzy controllers, and adaptive FLC.

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Author Contribution Declaration

Ashwini Yenegur: Conceptualization, Methodology, Software, Visualization, Investigation, Writing- Original draft preparation.
M.Sasikala: Data curation, Validation, Supervision, Resources, Writing - Review & Editing. Ashwini Yenegur and M.Sasikala :
Project administration, Supervision, Resources, Writing - Review & Editing.

Funding Declaration

"The author(s) received no financial support for the research, authorship, and/or publication of this article".

Data Availability Statement

The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request.
All data generated or analysed during this study are included in this published article.

The datasets generated and/or analysed during the current study are not publicly available but are available from the corresponding author on reasonable request.